PROBLEM 20

Heavy Particle on a Rotating Circumference

Original statement · Problems.pdf, p. 10

Rotating Curve & Energy Conservation

A heavy particle is moving along a smooth circumference of radius R and center O. One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity ω about it. The position of the particle with respect to the circumference is described by the angle θ between the particle and the lowest point of the circumference.

(a) Discuss the conservation of mechanical energy in the moving reference frame and compute its value.

(b) Compute the reaction of the circumference in the direction normal to its plane, as a function of θ.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m slides smoothly along a circular wire of radius R rotating with constant angular velocity ω about its vertical diameter.
- **CDOFs:** 1 degree of freedom, described by angle $\theta(t)$ from the lowest point of the circumference.
- **Hypotheses:** Frictionless wire, steady rotation $\boldsymbol{\omega} = \omega \mathbf{k}$.
- **Parameters:** $m > 0$ [kg], $R > 0$ [m], $\omega > 0$ [rad/s], $g > 0$ [m/s^2].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}\}$.
- **Rotating Frame:** $\mathcal{F}_1 = \{O; \mathbf{e}_\rho, \mathbf{e}_\phi, \mathbf{k}\}$ rotating with angular velocity $\boldsymbol{\omega}_{1/0} = \omega \mathbf{k}$.
- **Polar Basis on the Wire:** $\mathcal{B}_w = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$ where:

$$\mathbf{r} = R\sin\theta\,\mathbf{e}_\rho - R\cos\theta\,\mathbf{k} = R\mathbf{e}_r$$

$$\mathbf{e}_\theta = \cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}$$

- **Relative Velocity in $\mathcal{F}_1$:**

$$\mathbf{v}' = R\dot{\theta}\mathbf{e}_\theta$$

- **Change of basis $\mathcal{B}_0 \to \mathcal{B}_1$ (rotating frame).** With the wire plane at azimuth ωt, $\mathbf{e}_\rho = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$, $\mathbf{e}_\phi = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Change of basis $\mathcal{B}_1 \to \mathcal{B}_w$ (wire basis).** Since $\mathbf{e}_r \times \mathbf{e}_\theta = -\mathbf{e}_\phi$, the right-handed wire triple is $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_n\}$ with $\mathbf{e}_n = -\mathbf{e}_\phi$ (the ordering $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$ listed above has determinant -1; only the sign of the third vector is affected, and N_ϕ below is defined along $\mathbf{e}_\phi$). The columns are the components in $\{\mathbf{e}_\rho, \mathbf{e}_\phi, \mathbf{k}\}$:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_1R_w] = 1, \qquad [{}_1R_w][{}_1R_w]^T = I$$

- Determinants and products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Conservation of Mechanical Energy in the Moving Frame

Pedagogical Justification: In the rotating frame $\mathcal{F}_1$, fictitious forces appear: centrifugal force $\mathbf{F}_{cf} = m\omega^2\rho\mathbf{e}_\rho$ and Coriolis force $\mathbf{F}_{Cor} = -2m\boldsymbol{\omega} \times \mathbf{v}'$.

1. The centrifugal force is conservative, derived from potential $V_{cf} = -\frac{1}{2}m\omega^2\rho^2 = -\frac{1}{2}m\omega^2 R^2\sin^2\theta$.

2. The Coriolis force is always perpendicular to velocity ($\mathbf{F}_{Cor} \cdot \mathbf{v}' = 0$), doing strictly zero power.

3. The constraint reaction $\mathbf{N}$ is perpendicular to the wire ($\mathbf{N} \cdot \mathbf{e}_\theta = 0$), so $W_{\mathbf{N}} = 0$.

Therefore, the effective mechanical energy E' in the rotating frame is conserved.

Expressions for energies:

$$T' = \frac{1}{2}m|\mathbf{v}'|^2 = \frac{1}{2}mR^2\dot{\theta}^2$$

$$V_g(\theta) = -mgR\cos\theta \quad (\text{with reference datum } V_g(\pi/2) = 0)$$

$$V_{cf}(\theta) = -\frac{1}{2}m\omega^2 R^2\sin^2\theta$$

Total effective mechanical energy:

$$E' = T' + V_g + V_{cf} = \frac{1}{2}mR^2\dot{\theta}^2 - mgR\cos\theta - \frac{1}{2}m\omega^2R^2\sin^2\theta = \text{constant}$$

Part (b): Reaction Normal to the Plane of the Circumference

> **Pedagogical Justification:** The unit vector normal to the plane of the circumference is $\mathbf{e}_\phi$. Applying Newton's second law in the rotating frame along $\mathbf{e}_\phi$ isolates the out-of-plane reaction force N_ϕ.

Dynamic equilibrium in $\mathcal{F}_1$:

$$m\mathbf{a}' = \mathbf{F}_g + \mathbf{F}_{cf} + \mathbf{F}_{Cor} + \mathbf{N}$$

Since the particle is constrained to the planar wire, $a'_\phi = 0$.

Project forces along $\mathbf{e}_\phi$:

1. Gravity: $\mathbf{F}_g \cdot \mathbf{e}_\phi = (-mg\mathbf{k}) \cdot \mathbf{e}_\phi = 0$.

2. Centrifugal: $\mathbf{F}_{cf} \cdot \mathbf{e}_\phi = (m\omega^2 R\sin\theta\mathbf{e}_\rho) \cdot \mathbf{e}_\phi = 0$.

3. Coriolis:

$$\mathbf{F}_{Cor} = -2m(\omega\mathbf{k}) \times (R\dot{\theta}\mathbf{e}_\theta) = -2m\omega R\dot{\theta}(\mathbf{k} \times (\cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}))$$

Since $\mathbf{k} \times \mathbf{e}_\rho = \mathbf{e}_\phi$ and $\mathbf{k} \times \mathbf{k} = \mathbf{0}$:

$$\mathbf{F}_{Cor} = -2m\omega R\dot{\theta}\cos\theta\,\mathbf{e}_\phi$$

4. Reaction: $\mathbf{N} \cdot \mathbf{e}_\phi = N_\phi$.

Equating components:

$$0 = -2m\omega R\dot{\theta}\cos\theta + N_\phi \implies N_\phi = 2m\omega R\dot{\theta}\cos\theta$$

Using energy conservation to express $\dot{\theta}$ as a function of θ:

$$\dot{\theta} = \pm\sqrt{\frac{2}{mR^2}\left(E' + mgR\cos\theta + \frac{1}{2}m\omega^2R^2\sin^2\theta\right)}$$

$$N_\phi(\theta) = \pm 2m\omega\cos\theta\sqrt{\frac{2}{m}\left(E' + mgR\cos\theta + \frac{1}{2}m\omega^2R^2\sin^2\theta\right)}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- The out-of-plane reaction N_ϕ is purely generated by the Coriolis acceleration: to keep the particle moving along the rotating meridian, the wire must exert a lateral force balancing the Coriolis effect.
- At the equator ($\theta = \pi/2$), $\cos(\pi/2) = 0 \implies N_\phi = 0$. At this point, the velocity $\mathbf{v}'$ is parallel to $\boldsymbol{\omega}$, so the cross product $\boldsymbol{\omega} \times \mathbf{v}'$ vanishes.
- **Dimensional check:** $[N_\phi] = [m\,\omega\,R\,\dot{\theta}] = \mathrm{kg}\cdot\mathrm{s}^{-1}\cdot\mathrm{m}\cdot\mathrm{s}^{-1} = \mathrm{N}$; in E' each of $\frac{1}{2}mR^2\dot{\theta}^2$, $mgR\cos\theta$ and $\frac{1}{2}m\omega^2R^2\sin^2\theta$ has units of joule.
- **Limit $\omega \to 0$:** the wire is at rest, $N_\phi \to 0$ and $E' \to \frac{1}{2}mR^2\dot{\theta}^2 - mgR\cos\theta$, the energy of the simple pendulum.
- **Independent check of N_ϕ (sympy, inertial frame):** differentiating twice $\mathbf{r} = R\sin\theta\,\mathbf{e}_\rho(\omega t) - R\cos\theta\,\mathbf{k}$ in the inertial frame gives $\mathbf{a}\cdot\mathbf{e}_\phi = 2\omega R\dot{\theta}\cos\theta$; since only N_ϕ acts along $\mathbf{e}_\phi$, $N_\phi = 2m\omega R\dot{\theta}\cos\theta$, identical to the rotating-frame result.

- **Numerical check of E' (RK4, step 10^{-3}, $R = 1$ m, $\omega = 1.3$ rad/s, $g = 9.81$ m/s^2):** integrating $\ddot{\theta} = -(g/R)\sin\theta + \omega^2 \sin\theta\cos\theta$ for 20 s from $\theta = 0.4$, $\dot{\theta} = 0.3$, E'/m stays at -9.118750 with a drift of 9×10^{-14}.

Source: Problems.pdf (Page 10)